

Akash Karthik

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SUMMARY

Software engineer who delivers end-to-end features with **ownership**, clear **docs**, and tested code. Strong in Python and TypeScript for **APIs**, small **web apps**, and **data/tooling**. Comfortable with **HTTP/REST**, input validation, auth basics (sessions/JWT), **CI** (GitHub Actions), and **structured logging**. I thrive in **stand-ups/sprint planning**, ship via **PRs**, and keep quality high with regression checks.

EDUCATION

San Jose State University

B.S., Computer Science

Relevant coursework: Data Structures & Algorithms; Discrete Math; Linear Algebra; Embedded Systems

San Jose, CA

Expected Dec 2026

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java

Web & APIs: Flask/FastAPI, Node.js/Express, REST/JSON, OpenAPI (basic), Postman

Networking & Security: HTTP, TLS basics, cookies/sessions, JWT, input validation, rate limiting (basic), structured logging

Data & DB: pandas/NumPy, CSV/JSON pipelines, SQLite, SQL Server, schema design, seed/export scripts

DevOps: GitHub Actions CI/CD, Docker, reproducible envs, lint/test gates

Quality: pytest/Jest unit & regression tests, quick dashboards/plots for verification

Tools: Git/GitHub (branches/PRs/code reviews), VS Code, Figma (wireframes)

EXPERIENCE

Zip Intelligence

Sep 2024 -- Oct 2024

Software Development Intern — web tooling

Tracy, CA

- Implemented frontend **validation** and clearer error states; paired with engineers to harden **REST** endpoints (status codes, edge cases).
- Added SQL Server update scripts and **structured logs** to speed repro and triage; shipped via **PRs** with setup notes and change docs.
- Participated in **stand-ups/sprint planning**; took **ownership** of a small feature from ticket to verified release.

Spartan Racing, Formula SAE

Aug 2024 -- Present

Software/Controls — data and tooling

San Jose, CA

- Built Python utilities for **CSV ingestion** and quick plots to make telemetry reviews self-serve; standardized file layouts and checks.
- Developed a **closed-loop simulation harness** that replays logs against controller code, runs parameter sweeps, and auto-generates plots with pass/fail checks for regression.

Sailfish

Jan 2024 -- Aug 2024

Data Analyst — reporting dashboards

Tracy, CA

- Designed structured dashboards and weekly summaries; streamlined data entry; **32%** uplift reported in tracked metrics.
- Authored light **runbooks/docs** (inputs, queries, and QA checks) to keep reports maintainable.

PROJECTS

Tickr — AI-coached paper trading game

TypeScript/JavaScript | React | flat-file datastore | UX flows

- Scenario-driven learning game (e.g., historic events like the GME squeeze): users chat with an **AI coach**, then make “higher/lower” decisions and see outcomes.
- Built **registration/sign-in** flow, core game screens, and history view; wrote quest/prompt logic and basic state management; mobile-responsive layout.

Scent Showdown

Flask | HTML/CSS/JS | SQLite | Deploy

- Lightweight app with **matchup UI** and leaderboard; routes/templates, compact schema, input validation, and an admin view.
- Exposed JSON endpoints for data; seed/export scripts and a clear README for maintainers.

Educational Trading Bot

Python | APIs | pandas | CI

- EMA(9/21)+sentiment paper-trading loop with structured configs/logs for reproducible runs.
- ETL-style pipeline: ingest market/news, transform features, simulate trades, and output metrics/plots.